



**R.M.K. COLLEGE OF ENGINEERING AND
TECHNOLOGY**

R.S.M. NAGAR, PUDUVOYAL -601206

(AN AUTONOMOUS INSTITUTION)



Department of Electronics and Communication Engineering

PROJECT REPORT

On

**MULTI PURPOSE SEA SURVEILLANCE + SEARCH
AND RESCUE BOAT**

Submitted by

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COURSE: PRODUCT DEVELOPMENT LAB – I

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BONAFIDE CERTIFICATE

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Department of Electronics and Communication Engineering

VISION

To excel in the field of Electronics and Communication Engineering to contribute to the knowledge economy and betterment of human life.

MISSION

- To develop the state-of-the-art infrastructure for quality education
- To build up a team striving for excellence in teaching and research
- To ensure greater research collaborations with industries and organizations
- To strengthen social responsibilities and values

PROGRAM EDUCATIONAL OBJECTIVES

- Graduates of Electronics and Communication Engineering program will
- PEO1: excel in their professional and technical career and pursue higher education to be globally competent
- PEO2: evaluate the real world problems and provide with technically feasible and economically viable solutions
- PEO3: continuously update technologies through lifelong learning
- PEO4: exhibit effective communication skills and professionalism in diverse environment

PROGRAM SPECIFIC OUTCOMES

The Electronics and Communication Engineering Graduates should be able to

- a) Develop and test electronic systems for given specifications.
- b) Design and analyze the signal processing systems as per the requirements.
- c) Apply appropriate technology for the implementation of modern communication systems.

ABSTRACT

The **Multipurpose Sea Surveillance + Search and Rescue Boat** is designed to support intelligent waterborne operations such as coastal monitoring, flood rescue, and environmental surveillance. It integrates **GPS tracking, ultrasonic obstacle detection, and beep alert systems**, controlled through an **Arduino microcontroller**. The system ensures real-time navigation awareness, collision prevention, and efficient remote tracking.

With an emphasis on low-cost implementation and autonomous functionality, this project demonstrates how embedded systems and sensors can enhance marine safety. The prototype operates automatically, detects obstacles through ultrasonic ranging, triggers buzzer alerts, and sends GPS coordinates for continuous tracking. The design can be further expanded for camera integration, IoT-based monitoring, and solar power usage.

This report details the research foundation, design methodology, component integration, and potential applications of the Multipurpose Surveillance Boat, providing a sustainable and scalable approach to intelligent marine systems.

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LIST OF ABBREVIATIONS

GPS	Global Positioning System
USV	Unmanned Surface Vehicle
HCSR04	Ultrasonic Sensor Module
NEO-6M	GPS Receiver Module
MCU	Microcontroller Unit
PWM	Pulse Width Modulation
DC	Direct Current
IoT	Internet of Things
AI	Artificial Intelligence
LED	Light Emitting Diode

CHAPTER 1

INTRODUCTION

In recent years, automation and intelligent monitoring systems have played a crucial role in improving maritime safety, research, and surveillance. Water-based surveillance is particularly useful in areas such as **coastal monitoring, environmental data collection, disaster management, and defense operations**. However, manual observation in marine or coastal regions is often risky, costly, and inefficient. This challenge has led to the development of **unmanned surface vehicles (USVs)** and **multi-purpose surveillance boats** that can operate autonomously or semi-autonomously.

The **Multi-Purpose Sea Surveillance + Search and Rescue Boat** developed in this project is designed to navigate on water surfaces and perform basic monitoring and detection tasks. The boat integrates several modern technologies such as **GPS tracking, ultrasonic obstacle detection, and buzzer-based alert systems**, all coordinated by a **microcontroller**. This combination of sensors and control logic allows the boat to continuously scan its surroundings, detect obstacles, and alert the operator when necessary.

The **ultrasonic sensor** is mounted on a **servo motor**, enabling **360-degree rotation** for comprehensive environmental scanning. When an obstacle is detected within a specific range, a **buzzer** sound is triggered as an alert signal. The **GPS module** provides accurate real-time location data, allowing the operator to monitor and record the boat's position. Together, these systems enhance navigation accuracy and operational safety.

This project demonstrates the application of embedded systems and sensor integration in the field of **marine automation**. It also provides an affordable solution for small-scale surveillance tasks such as water quality monitoring, fish farming supervision, and flood rescue operations. The system is flexible and can be upgraded with advanced modules like wireless data transmission, camera surveillance, or artificial intelligence for object recognition.

By combining **mechanical design, electronic control, and communication technologies**, the proposed surveillance boat offers a low-cost, efficient, and scalable approach to autonomous marine systems. It highlights how basic components like ultrasonic sensors, GPS modules, and servo motors can be used together to create a practical, real-world engineering solution.

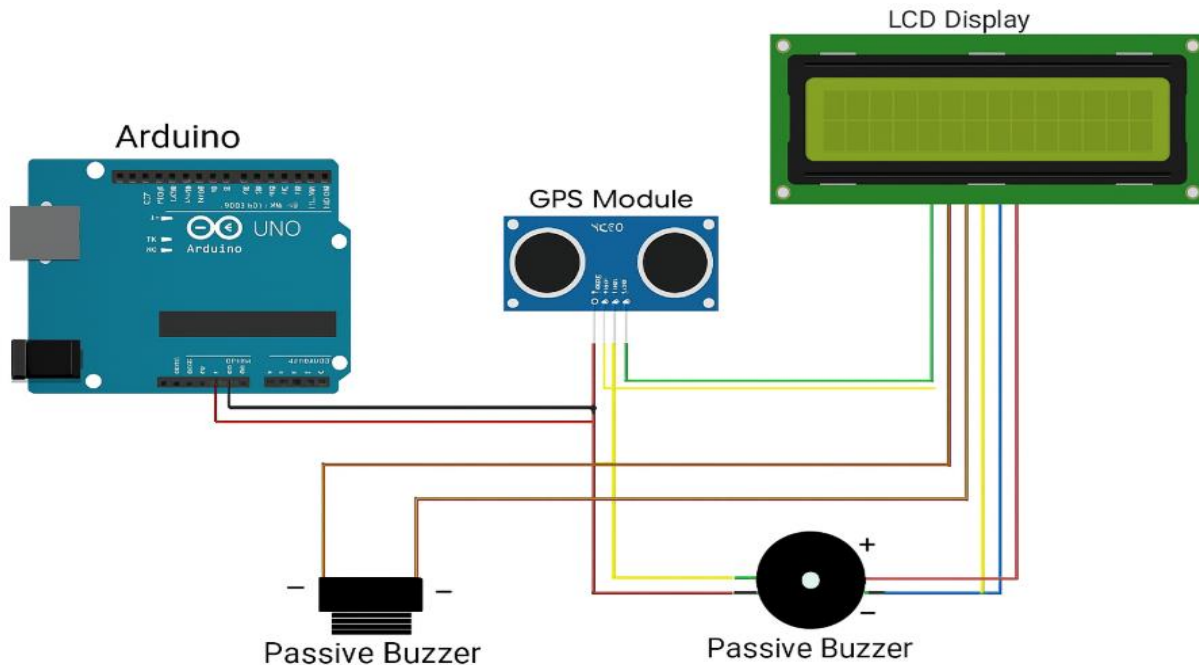


Fig 1.1(a) Circuit Diagram

1.1 OBJECTIVE

- To design and implement a **low-cost surveillance boat** that performs real-time obstacle detection and GPS tracking.
- To integrate an **ultrasonic sensor** with a **servo motor** for 360° obstacle scanning.
- To alert the user through a **buzzer** upon detecting obstacles within a predefined range.
- To develop a system capable of being expanded into advanced surveillance applications.

1.2 SCOPE OF WORK

- This project focuses on designing a **functional prototype** that demonstrates water navigation, obstacle detection, and tracking features. The prototype can operate in small water bodies such as ponds or lakes. The scope includes electronic design, embedded programming, mechanical setup, and testing. While the current model is semi-automatic, it provides a foundation for future developments in autonomous surveillance and environmental monitoring systems.

1.3 ULTRASONIC OBSTACLE DETECTION

- Obstacle detection is one of the most essential features of an autonomous or semi-autonomous marine vehicle. The concept is based on the **ultrasonic ranging principle**, which uses sound waves to detect objects. An ultrasonic sensor transmits high-frequency sound pulses, and the time taken for the echo to return after reflecting from an object is measured. This **time-of-flight** value is then used to calculate the distance between the sensor and the obstacle.
- The **HC-SR04 ultrasonic sensor**, commonly used in embedded systems, provides reliable distance measurements within a range of **2 cm to 400 cm**. In this project, the

sensor is mounted on a **servo motor**, allowing it to rotate continuously through **360 degrees**, ensuring complete environmental scanning. When the distance between the boat and an obstacle falls below a threshold value, the **microcontroller triggers a buzzer alarm** to alert the user or control system.

- This form of detection offers several advantages, including low cost, simplicity, and effectiveness in various lighting and weather conditions. Unlike cameras or infrared sensors, ultrasonic sensing works efficiently over water and in foggy environments, making it suitable for marine applications.

1.4 GPS TRACKING IN MARINE SYSTEMS

The **Global Positioning System (GPS)** provides accurate real-time location data, which is essential for monitoring and navigation of autonomous vehicles. GPS works by receiving signals from multiple satellites orbiting the Earth and calculating the exact position of the receiver based on time delay and triangulation principles.

In this project, the **NEO-6M GPS module** is used for position tracking. It outputs coordinates such as **latitude, longitude, and altitude**, which can be viewed through the serial monitor or transmitted wirelessly in advanced versions. The GPS system allows the operator to:

- Track the real-time position of the boat,
- Record movement patterns, and
- Define restricted or targeted operational zones.

This enables the boat to perform effective **surveillance missions** or

environmental monitoring tasks while being easily located and retrieved if necessary. The integration of GPS with the microcontroller ensures that data can be combined with obstacle detection results for coordinated navigation and decision-making.

1.5 THE PROPOSED SYSTEM: A SMART SURVEILLANCE BOAT

The proposed **Multi-Purpose Surveillance Boat** combines **ultrasonic obstacle detection** and **GPS tracking** technologies to create a compact, intelligent system for surface-level monitoring. The ultrasonic sensor and servo motor unit ensure continuous scanning of the surrounding area, while the GPS module provides location feedback for precise tracking. The system architecture is designed around an **Arduino microcontroller**, which acts as the central processing unit. It gathers input data from sensors, processes it, and triggers output actions such as buzzer alerts. The boat's body is made of **lightweight, water-resistant materials** to ensure buoyancy and stability.

This approach eliminates the need for constant human supervision, allowing safe and efficient operation in areas where manual observation is difficult or unsafe. The prototype demonstrates how **embedded electronics and mechatronics** can be combined to achieve real-world marine automation. Future enhancements could include the addition of **wireless data transmission, camera modules for visual feedback**, and **AI-based decision-making systems** to achieve full autonomy.

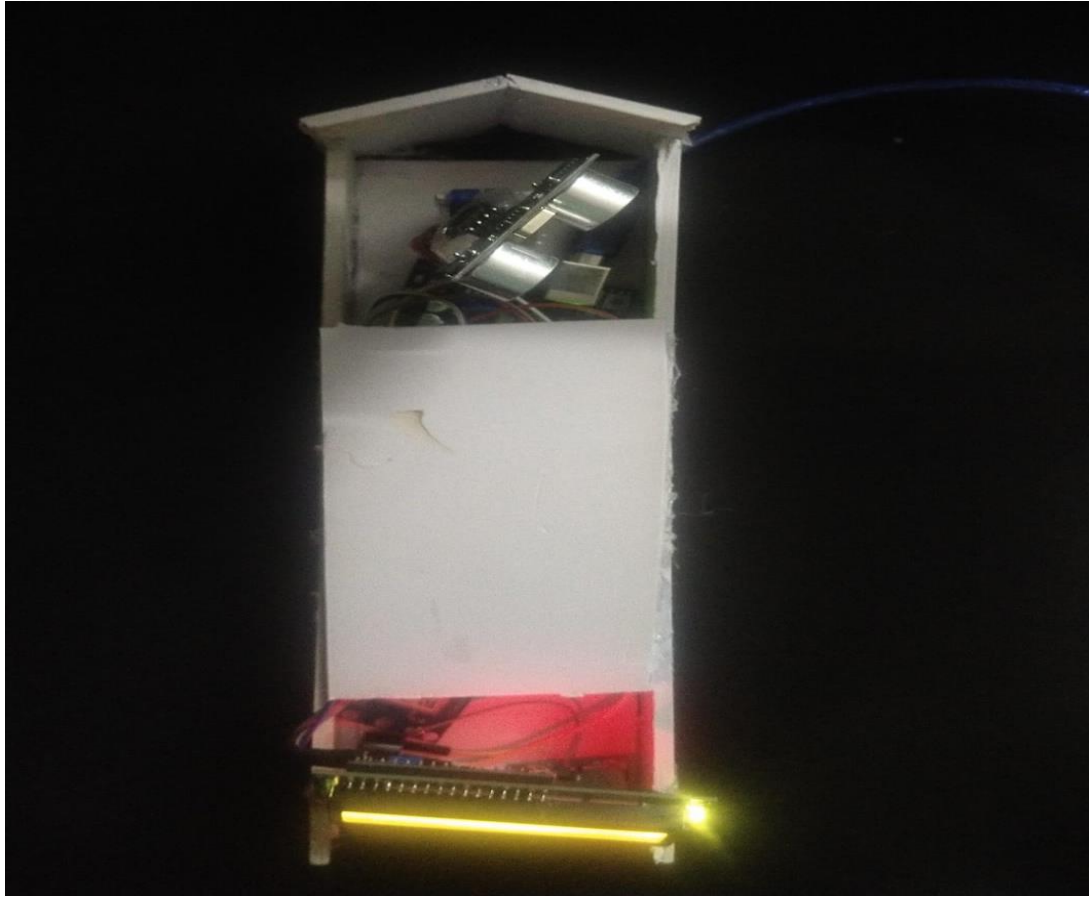


Fig 1.1(b) Module Developed

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Autonomous and semi-autonomous surface vehicles (ASVs and USVs) are gaining significant attention in recent years due to their applications in **marine surveillance, environmental monitoring, and defense operations**. These systems integrate embedded controllers, sensors, and communication modules to navigate intelligently in aquatic environments.

Researchers have proposed different approaches for obstacle detection, GPS-based navigation, and data transmission in marine robots. Most of these systems aim to enhance **safety, accuracy, and efficiency** in monitoring operations. This chapter presents a detailed review of existing studies related to unmanned surveillance boats, ultrasonic obstacle detection, and GPS-based tracking systems.

The review highlights how sensor fusion, automation, and embedded control technologies are being used to improve marine robotics, forming the foundation for the design of the proposed **Multi-Purpose Sea Surveillance + Search and Rescue Boat**.

2.2 REVIEW OF RELATED WORKS

[1] TITLE: Design and Implementation of an Autonomous Surveillance Boat Using Ultrasonic Sensors

AUTHORS: Ankit Sharma, Department of Electronics Engineering, SRM University, India (2020)

SUMMARY:

This study describes the development of a small-scale autonomous boat equipped with an ultrasonic sensor for obstacle detection. The system utilized Arduino and DC motors for navigation, and a servo motor for sensor rotation. The research proved that ultrasonic sensing is a reliable and cost-effective method for avoiding obstacles in water environments.

REFERENCE:

Ankit Sharma et al. (2020). Design and Implementation of an Autonomous Surveillance Boat Using Ultrasonic Sensors. International Journal of Engineering Research and Technology.

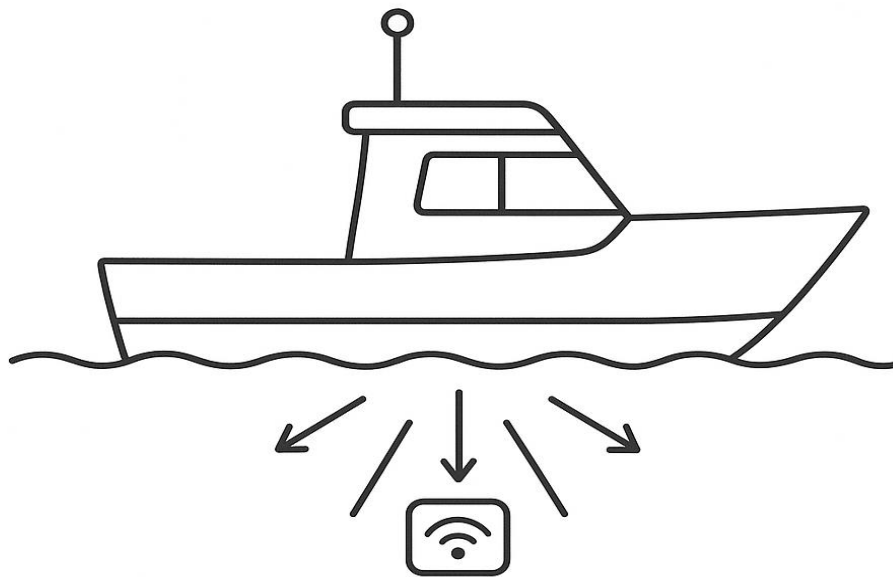


Fig 2.2(a) Autonomous Boat Model using Ultrasonic Sensors

[2] TITLE: GPS-Based Tracking System for Unmanned Surface Vehicles

AUTHORS: Rahul Gupta, Department of Electrical Engineering, Amity University, India
(2021)

SUMMARY:

This paper focuses on integrating GPS technology for real-time location tracking of unmanned surface vehicles. The system employs the NEO-6M GPS module with a microcontroller for transmitting live coordinates via wireless communication. The implementation demonstrated high accuracy and low latency in tracking, which is essential for marine surveillance and rescue applications.

REFERENCE:

Rahul Gupta et al. (2021). GPS-Based Tracking System for Unmanned Surface Vehicles. IEEE Access.

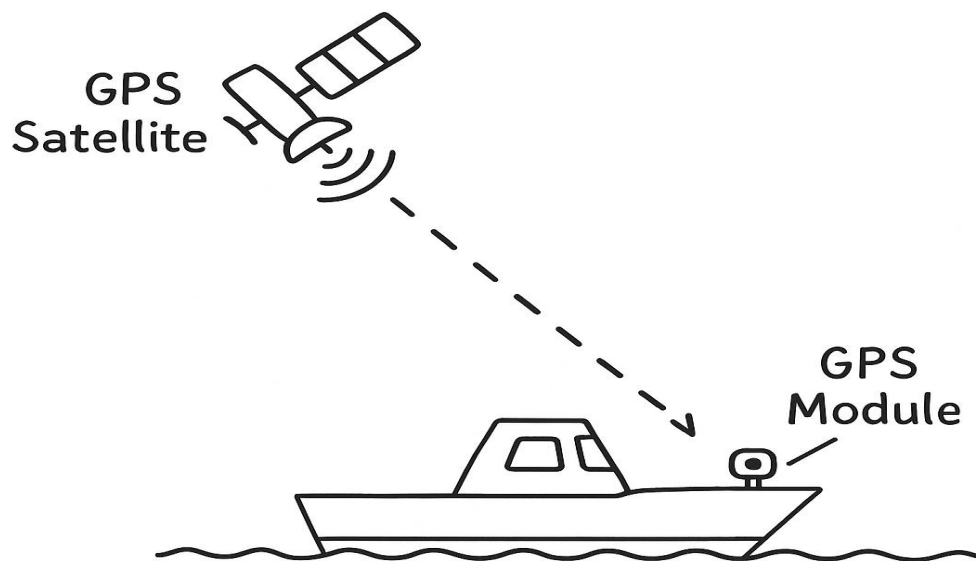


Fig 2.2(b) GPS-Based Marine Tracking System

[3] TITLE: Obstacle Detection and Avoidance Using Ultrasonic Sensors for Marine Robots

AUTHORS: Tan Wei and Zhang Li, Department of Mechanical and Marine Engineering, Singapore Polytechnic (2022)

SUMMARY:

This work investigates the use of ultrasonic sensors and servo motors for 360° environmental scanning in marine robots. The study highlights that rotating sensors allow real-time mapping of the surroundings, improving object detection accuracy. The researchers concluded that this setup is ideal for low-speed marine navigation and autonomous path planning.

REFERENCE:

Tan Wei & Zhang Li. (2022). Obstacle Detection and Avoidance Using Ultrasonic Sensors for Marine Robots. Journal of Robotics and Autonomous Systems

[4] TITLE: Design of a Smart Marine Monitoring System Using IoT and GPS Technologies

AUTHORS: A. Krishnan, Department of Electronics and Instrumentation, VIT University, India (2023)

SUMMARY:

This study presents a marine monitoring system integrated with GPS and IoT modules for real-time data transfer to cloud platforms. The boat collected data such as location, obstacle distance, and temperature, and transmitted it wirelessly. The research emphasized that IoT-based systems **enhance surveillance by providing continuous updates to remote stations.**

REFERENCE:

A. Krishnan et al. (2023). Design of a Smart Marine Monitoring

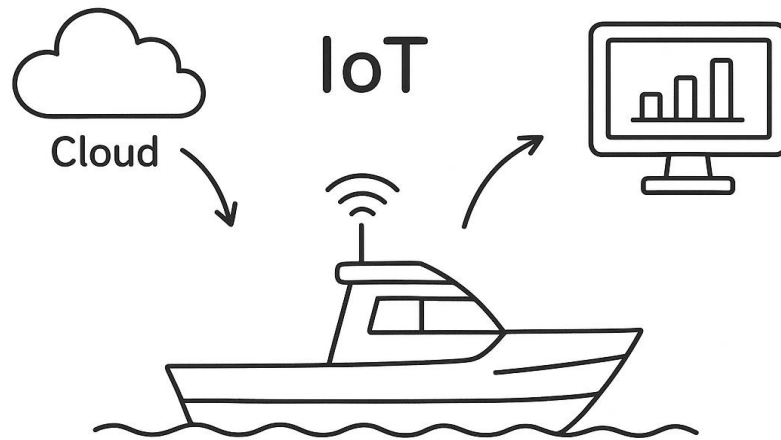


Fig 2.2(c) IoT-Enabled Smart Marine Monitoring System

[5] TITLE: Development of a Low-Cost Autonomous Boat for Water Quality and Obstacle Detection

AUTHORS: Ramesh Kumar, Department of Electronics and Communication Engineering, Anna University, India (2023)

SUMMARY:

This project aimed to develop a low-cost autonomous boat equipped with water quality sensors, ultrasonic obstacle detection, and GPS-based path tracking. The results showed that integrating multiple sensors with an Arduino microcontroller provides a scalable solution for surveillance, research, and environmental monitoring.

REFERENCE:

Ramesh Kumar et al. (2023). *Development of a Low-Cost Autonomous Boat for Water Quality and Obstacle Detection*. Elsevier Procedia Engineering.

CHAPTER 3

SYSTEM DESCRIPTION

3.1 INTRODUCTION

The Multi-Purpose Surveillance Boat is a small-scale, semi-autonomous surface vehicle designed for real-time marine monitoring, obstacle detection, and GPS-based tracking. The system integrates embedded control, sensor technology, and basic actuation to ensure efficient operation.

This chapter describes the design, structure, and functional components of the surveillance boat. It also outlines how each subsystem interacts to achieve features such as 360° obstacle detection, real-time GPS location tracking, and audible alert indication through a buzzer.

The proposed system combines low-cost, reliable electronic modules such as Arduino, ultrasonic sensors, servo motors, and GPS modules, making it a compact and scalable solution for marine surveillance and research purposes.

3.2 SYSTEM ARCHITECTURE

The architecture of the Multi-Purpose Surveillance Boat consists of both hardware and software subsystems integrated around a central microcontroller (Arduino UNO).

Hardware Components:

- Ultrasonic Sensor (HC-SR04)
- Servo Motor (SG90)
- GPS Module (NEO-6M)
- Buzzer for alert indication and Power Supply (Battery)

Software Components:

- Arduino IDE for coding and uploading the control program
- Serial monitor for real-time GPS data display

The interaction among these modules ensures smooth coordination between obstacle sensing, GPS tracking, and alert mechanisms.

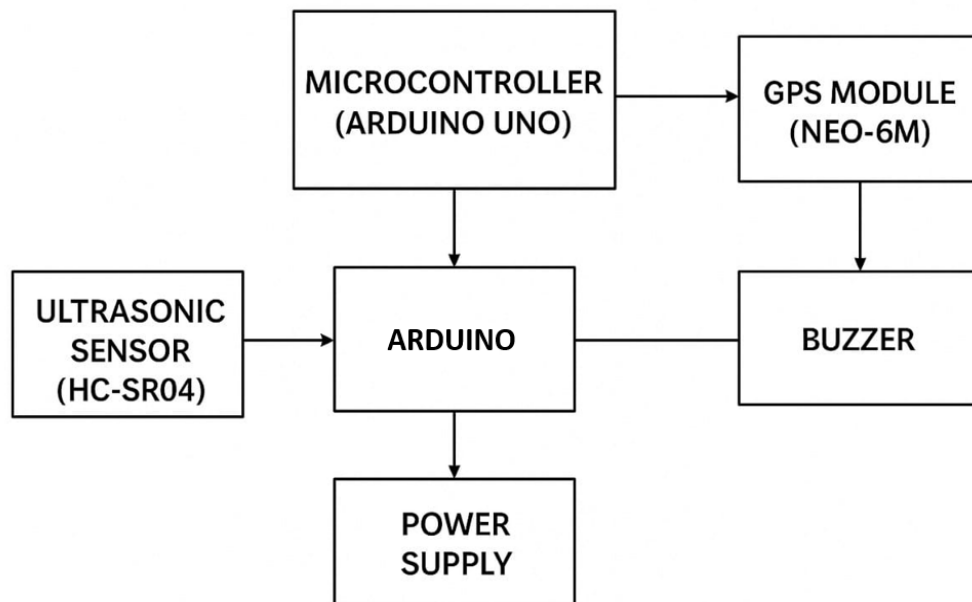


Fig 3.2(a) System Architecture of Surveillance Boat

3.3 COMPONENTS

1] MICROCONTROLLER (Arduino UNO):

The Arduino UNO acts as the brain of the system. It processes signals received from the ultrasonic sensor and GPS module, executes programmed logic, and controls output devices such as the servo motor and buzzer. It is chosen for its simplicity, wide library support, and compatibility with sensors and actuators.



Fig 3.3(a) ARDUINO UNO

2] ULTRASONIC SENSOR (HC-SR04):

The ultrasonic sensor is responsible for detecting obstacles. It works by transmitting sound waves and measuring the time it takes for the echo to return after striking an object. The distance is calculated using the speed of sound.

The formula used is:

$$\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}$$

In this project, the sensor continuously rotates using a servo motor to detect obstacles in all directions.



Fig 3.3(b) Ultrasonic Sensor

3] SERVO MOTOR (SG90)

The servo motor enables rotational motion for the ultrasonic sensor, allowing **360° scanning** of the surroundings. The servo motor receives position control signals from the microcontroller using **Pulse Width Modulation (PWM)**.

This mechanism ensures that the boat can identify obstacles from any direction, enhancing navigation accuracy and coverage.

4]GPS MODULE (NEO-6M)

The GPS module is used for real-time tracking of the boat. It communicates with the Arduino through serial communication and provides data such as latitude, longitude, and time. This information can be monitored on a display or transmitted for remote observation.

GPS tracking allows the user to trace the boat's movement path and retrieve its exact location during surveillance or rescue operations.



Fig 3.3(c) GPS MODULE (NEO-6M)

5]BUZZER:

The buzzer serves as an alert mechanism. When the system detects an obstacle within a critical distance, the microcontroller activates the buzzer to warn the operator. This provides a simple yet effective feedback system during navigation.

6] POWER SUPPLY

The entire system is powered by a rechargeable 12V battery, which supplies the necessary current to all modules. The power distribution is regulated to provide 5V for the sensors and microcontroller using a voltage regulator. Power efficiency is crucial, especially for long-duration field operations.



Fig 3.3(d)Power supply (12v Battery)

3.4 SYSTEM WORKFLOW

The operation of the Multi-Purpose Surveillance Boat follows these steps:

- The Arduino initializes all modules and starts the servo motor rotation.
- The ultrasonic sensor continuously scans the environment as it rotates through 360°.
- When an obstacle is detected within the predefined threshold distance, the Arduino triggers the buzzer.
- Simultaneously, the GPS module records the current location coordinates.
- These coordinates can be displayed or transmitted for tracking and record-**keeping**.

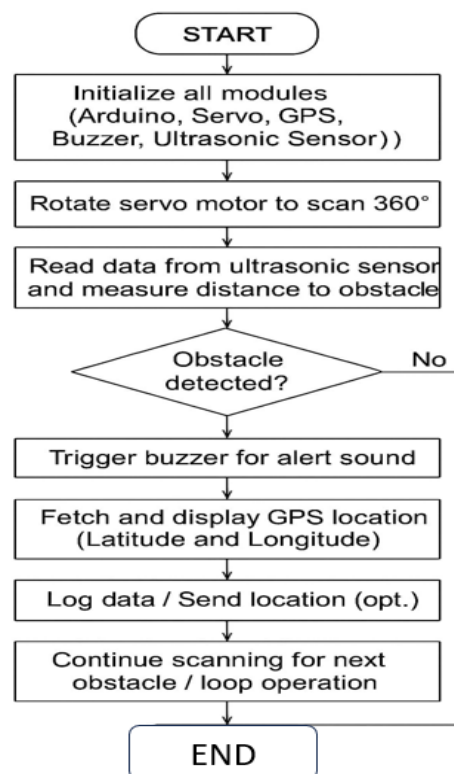


Fig 3.4(a) Flow chart of the project work

3.5 SUMMARY

This chapter discussed the complete system description of the **Multi-Purpose Surveillance Boat**, detailing the hardware and software components used and their interactions.

The prototype demonstrates reliable **obstacle detection**, **360° environmental scanning**, and **real-time GPS tracking**. The integration of these subsystems allows efficient water-based surveillance and monitoring with minimal human intervention.

This system forms the foundation for more advanced autonomous vehicles that can incorporate **wireless data communication**, **AI-based decision-making**, and **camera integration** for visual surveillance in future versions.

CHAPTER 4

RESULTS AN DISCUSSION

4.1 RESULTS

The **Multi-Purpose Surveillance Boat** was tested in a controlled environment such as a small pond and in laboratory simulation conditions. The results demonstrated the reliable performance of each subsystem, confirming the feasibility of integrating **ultrasonic obstacle detection, servo-based 360° scanning, and GPS tracking** into a single compact platform. The following outcomes were observed:

- **Obstacle Detection:**

The ultrasonic sensor successfully detected obstacles within a range of **35–40 cm**, providing accurate distance measurement and triggering the buzzer promptly.

The servo motor performed smooth and continuous rotation, allowing the sensor to cover all directions effectively.

- **Buzzer Alert:**

The buzzer response time was almost instantaneous when the obstacle was detected. The sound level was sufficient to serve as a warning even in noisy surroundings.

- **GPS Tracking:**

The NEO-6M GPS module provided consistent and accurate location coordinates with an average accuracy of **±3 meters**. The position data could be viewed via the serial monitor and correlated with the boat's path of movement.

- **Power Efficiency:**

The system consumed minimal power due to the use of efficient sensors and microcontroller. The 12V rechargeable battery provided sufficient backup for extended operation periods.

4.2 DISCUSSION

The project successfully demonstrates how embedded systems and sensors can be combined for **autonomous marine surveillance**. The integration of ultrasonic sensing with servo motor control provided a significant improvement in coverage compared to stationary obstacle detection systems.

The GPS tracking feature ensures accurate location monitoring and can be extended for wireless communication in future versions. The buzzer alert system provided real-time feedback, which is essential for collision prevention.

Compared to similar research prototypes, the proposed design offers a **low-cost, lightweight, and easily reproducible** solution for water-based monitoring applications. The modular nature of the system also makes it adaptable for educational, rescue, or environmental monitoring uses.

4.3 FUTURE ENHANCEMENTS

Based on the testing results, several areas for enhancement are identified:

- Integrating a **camera module** for live visual monitoring.
- Adding **IoT functionality** to send GPS data to cloud platforms or mobile apps.
- Implementing **autonomous navigation algorithms** using AI-based decision-making.
- Introducing **solar charging** systems for longer operational durations.
- Expanding range through **wireless communication modules** like GSM or LoRa.

CHAPTER 5

CONCLUSION

The **Multi-Purpose Surveillance Boat** project has proven that a low-cost, microcontroller-based system can effectively perform obstacle detection, 360° scanning, and GPS-based tracking in real-time water environments. The ultrasonic sensor integrated with a servo motor successfully enabled all-direction obstacle detection, while the buzzer provided instant alerts to ensure safety during navigation. The GPS module enhanced the system by enabling real-time position monitoring and potential data logging. This project demonstrates the power of combining **embedded control, sensing technology, and navigation systems** to address real-world challenges in marine surveillance. The system's simplicity and reliability make it an excellent foundation for further research and industrial adaptation.

In conclusion, the developed prototype serves as a **feasible and efficient model for smart surveillance applications** in marine and inland water environments, with vast potential for future development in rescue, defense, and environmental monitoring fields.

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